Student Management System: Create a Java program that allows a user to register multiple students. Create a Student class with attributes such as student ID, name, age, department, and marks, and use a constructor to initialize each student object using data entered through Scanner. The age must be between 16 and 60, and each mark must be between 0 and 100; if the user enters an invalid value, use a while or do...while loop to display an error and allow them to retry until the input is valid. Use a for loop to enter marks for five subjects, calculate the total and average, and use if/else if statements to determine whether the student has passed or failed and assign a grade. Finally, display all the information stored in the student object.

Bank Account System: Create a Java program that simulates a simple bank account. Create a BankAccount class containing account number, account holder name, PIN, and balance, and initialize the object using a constructor with information entered by the user. The user must create a four-digit PIN and enter an initial balance greater than zero; use loops to allow retries when invalid information is entered. After creating the object, ask the user to enter their PIN and allow a maximum of three attempts using a for or while loop. If authentication succeeds, use a do...while loop to repeatedly display options for checking the balance, depositing, withdrawing, and exiting. Use if statements to ensure deposits and withdrawals are greater than zero and prevent withdrawals greater than the available balance.

Employee Payroll System: Create a Java program for managing employee salaries. Create an Employee class with employee ID, name, department, basic salary, hours worked, and overtime hours, and use a constructor to initialize employee objects from information entered by the user. Ask the user how many employees they want to register and use a for loop to create each employee object. Use while or do...while loops to validate that salary is greater than zero and working hours are within an acceptable range, allowing the user to retry whenever invalid information is entered. Use if/else statements to calculate overtime when an employee works more than 160 hours, calculate tax according to different salary ranges, and determine the employee's final net salary. After processing all employees, display their payroll information and identify the employee with the highest net salary.

Hotel Booking System: Create a Java program for managing hotel bookings. Create a HotelBooking class containing booking ID, customer name, room type, number of nights, number of guests, and total amount, and use a constructor to initialize each booking object using information provided by the user. The hotel should provide Standard, Deluxe, and Executive rooms with different prices. Use a do...while loop to ensure that the user selects a valid room type and use a while loop to validate that the number of nights is between 1 and 30 and the number of guests is within the allowed limit. Use if/else if statements to determine the room price and discount based on the number of nights. Use a for loop to allow several customers to make bookings, and display the booking details, original amount, discount, and final amount for every created object.

Product and Shopping System: Create a Java program that allows a customer to purchase products from a shop. Create a Product class containing product ID, product name, price, and available quantity, and initialize several product objects using constructors with data entered by the user. Use a for loop to create and display the products, then allow the customer to enter the ID of the product they want to purchase. Use a while loop to keep asking if the entered product ID does not exist, and use another loop to validate that the requested quantity is greater than zero and does not exceed the available stock. Use if/else if statements to calculate discounts based on the total purchase amount and reduce the stock after a successful purchase. Use a do...while loop to allow the customer to continue purchasing other products until they choose to stop, then display the total amount paid and the remaining stock.